

PPA Exposure Model

Challenge Notes — Anticipated Questions & Answers

This document prepares for the technical challenge of the PPA exposure model. It covers the seven most likely questions from the panel — traders, risk managers, IT, and internal audit — with structured answers that demonstrate both understanding of the model and awareness of its limitations.

Challenge 1

"Why Schwartz one-factor and not two-factor?"

The one-factor model captures the dominant risk driver: mean-reverting spot price dynamics with seasonality. The Schwartz-Smith two-factor extension adds a stochastic long-term equilibrium, which matters most for contracts beyond 5-7 years where structural shifts in the energy market can materially change the equilibrium price level.

For a 5-year PPA, the mean-reversion in our model pulls prices back to equilibrium fast enough that the one-factor limitation is manageable. With $\kappa = 1.5$, the half-life of a price shock is approximately $\ln(2)/1.5 \approx 0.46$ years — within 6 months, most of the shock has dissipated. The real limitation appears at the long end of the tenor: the forward curve flattens quickly because one factor cannot generate a rich term structure.

For the panel: "The one-factor model is deliberately chosen as a first iteration. It gives us a working prototype we can validate and deploy, with a clear upgrade path to Schwartz-Smith when we extend to longer tenors or need richer term structure dynamics."

Challenge 2

"How would you calibrate κ and σ in practice?"

For σ : take historical log-returns of the spot price (e.g., daily or weekly TTF or baseload power prices), compute the standard deviation, and annualize. If options are liquid enough, use implied volatility from power options — this gives the market's forward-looking view rather than backward-looking historical vol.

For κ : run a regression on historical log-prices. The OU process in discrete time is:

$$X(t+1) = \alpha + \beta \cdot X(t) + \varepsilon \text{ where } \beta = \exp(-\kappa \cdot \Delta t)$$

So you regress today's log-price on yesterday's, and $\kappa = -\ln(\beta)/\Delta t$. A β close to 1 means slow reversion (low κ), close to 0 means fast reversion. With daily data for European baseload power, κ typically lands between 1.0 and 3.0 — the default of 1.5 is mid-range.

For the panel: "In production, I'd calibrate κ and σ to historical spot data using OLS on the discretized OU process, and cross-check σ against option-implied vols where available. The sidebar lets you override both parameters for scenario analysis."

Challenge 3

"The capacity factor is fixed — but wind farms don't produce constantly."

This is the most intellectually interesting limitation. In reality for offshore wind, there is a **negative correlation** between volume and price: when wind is high, power output is high, but supply is also high, so the spot price

drops.

For the PPA exposure, this means: when prices drop (positive MtM for the generator), wind output tends to be higher than average, so the actual delivered volume is larger than our fixed capacity factor assumes. Conversely, when prices rise (negative MtM), output tends to be lower. The net effect: our model with fixed capacity factor **overestimates the positive exposure** because high-exposure scenarios coincide with lower actual output in reality.

Studies on wind-price correlation in European markets show correlation coefficients around -0.3 to -0.5. The impact on PFE is probably 10-20% overestimation — meaningful but not catastrophic, and crucially it errs on the **conservative side** for credit risk.

For the panel: "The fixed capacity factor is a conservative assumption for credit exposure because the volume-price correlation is negative. I'd flag this as the first extension to implement, using a correlated bivariate simulation of price and wind capacity factor."

Challenge 4

"Your MtM at $t=0$ is €4.1M — shouldn't a new PPA start at zero?"

Correct — a freshly struck PPA priced at-market would have $MtM \approx 0$ at inception. The non-zero initial MtM in our defaults reflects the gap between the fixed price (55 EUR/MWh) and the model-implied forward curve, which starts at the spot (50 EUR/MWh) and converges toward the long-run mean (~52-53 EUR/MWh). The PPA is in-the-money from day one.

This is a **parameter choice, not a bug**. The defaults deliberately create a scenario with meaningful positive initial exposure to demonstrate the full exposure profile. In production, a newly executed PPA would have the fixed price set at or near the prevailing forward curve level.

To demonstrate a fair-value PPA in the demo, set the fixed price equal to the average forward curve level (~52-53 with defaults). The initial MtM drops to near zero and the exposure profile becomes more symmetric.

For the panel: "The initial MtM reflects the difference between the fixed price and the forward curve. In production, a newly executed PPA would be priced at-market, giving near-zero initial MtM. The defaults illustrate a scenario with meaningful existing exposure."

Challenge 5

"The PFE peaks at 1.1 years then declines — shouldn't uncertainty grow over time?"

This is the classic "humped" exposure profile, and it's one of the most important features of the model to understand. Two opposing forces are at play:

1. **Price uncertainty grows with time** — the further out you look, the wider the distribution of possible prices. This pushes PFE up.
2. **Remaining contract value shrinks with time** — as delivery periods pass, there are fewer remaining cash flows to value. A PPA at month 58 only has 2 months of deliveries left, so even a massive price move produces a small MtM. This pulls PFE down.

The peak occurs where these two forces balance. Early in the contract, there is enough uncertainty to generate significant MtM moves AND enough remaining tenor to make those moves valuable. Late in the contract, the remaining value is too small for even large price moves to matter.

This is textbook for any swap-like instrument — interest rate swaps show the same shape. If the PFE didn't decline toward maturity, the model would be wrong. The peak timing (1.1 years with defaults) is driven mainly by κ : high mean-reversion pulls prices back quickly, so maximum divergence from equilibrium happens early before reversion kicks in. Lower κ would push the peak later.

For the panel: "The humped profile is expected for a swap-like contract. It reflects the trade-off between growing price uncertainty and declining remaining contract value. The peak timing is governed by the mean-reversion speed."

Challenge 6

"Monthly time steps — is that granular enough?"

Monthly is sufficient for this use case. The PPA settles on a monthly basis, so sub-monthly price dynamics don't directly affect the exposure profile — what matters is where the price is at each settlement date.

Where monthly **breaks down**:

- If modeling intraday margin calls (CCP context), you need daily or even intraday steps. If modeling power price spikes (which can occur and revert within hours), monthly steps smooth them out completely — the spike never appears. This is related to the "no jumps" limitation.

Going from monthly to daily would multiply computation time by ~21x (60 steps to 1,260) while providing negligible improvement in PFE accuracy for a monthly-settling PPA. The convergence analysis confirms stability at 10,000 paths with monthly steps.

For the panel: "Monthly resolution matches the PPA settlement frequency. Daily steps would increase computation by 20x without materially improving the exposure estimate. If we were modeling intraday margin requirements, we'd need higher frequency."

Challenge 7

"What about wrong-way risk?"

This is the most advanced challenge and the one that separates senior risk practitioners from juniors. Wrong-way risk means the counterparty is more likely to default precisely when your exposure to them is highest.

In energy markets, this is very real: imagine the PPA counterparty is a utility that buys power at the fixed price. If market prices drop significantly (your MtM goes very positive — they owe you a lot), their business is also under pressure — they're buying at above-market prices across their portfolio. Their credit quality deteriorates at exactly the moment your exposure peaks. The probability of default and the exposure are positively correlated.

Our model assumes they're independent — we compute PFE as a pure market risk metric, ignoring the credit dimension. In a full CVA framework, you'd model $PD(t)$ as a function of the exposure level or the underlying risk factor:

$$PD(t \mid S_t) = \text{base_PD} \times f(\text{exposure})$$

where f increases when exposure increases. This amplifies the expected loss beyond what the simple $PFE \times PD$ calculation gives you.

For a prototype, acknowledging wrong-way risk without modeling it is appropriate. It's a second-order effect that requires counterparty-specific credit modeling (CDS spreads, financial statements, structural models).

For the panel: "The model computes market risk exposure independently from credit quality. In practice, wrong-way risk means the counterparty is more likely to default when our exposure is highest. I'd address this through the CVA framework — modeling PD as a function of the underlying risk factor. For the PFE application, the current approach is conservative because it doesn't net down exposure for low-default-probability scenarios."

Priority for the panel

The questions most likely to arise, in order of probability:

Priority	Challenge	Who asks
1	Humped PFE profile (#5)	Brendan / Risk Manager
2	Initial MtM not zero (#4)	Trader / Audit
3	Why one-factor (#1)	Brendan / Quant

4	Volume-price correlation (#3)	Trader / Risk Manager
5	Calibration (#2)	Quant / Audit
6	Wrong-way risk (#7)	Senior Risk / Audit
7	Time step granularity (#6)	IT / Quant

If you bring up wrong-way risk (#7) proactively — even briefly — it demonstrates depth that goes beyond the model itself. It shows you think about the interaction between market risk and credit risk, which is exactly what the Liquidity and Quantitative Risk Manager role requires.
